BANA 7042: Final Exam

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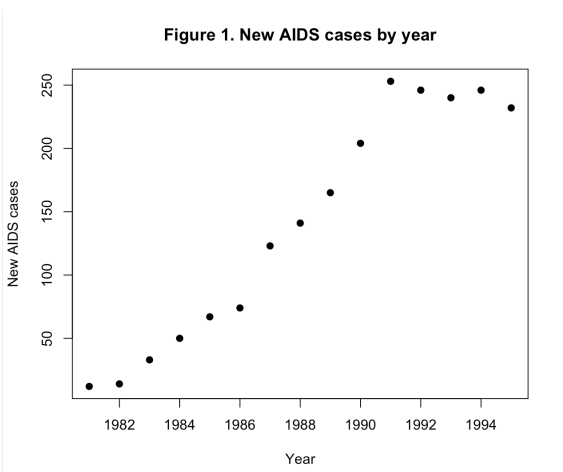
Question 1 – AIDS Cases (Poisson Regression)

(a) Plot and Comment

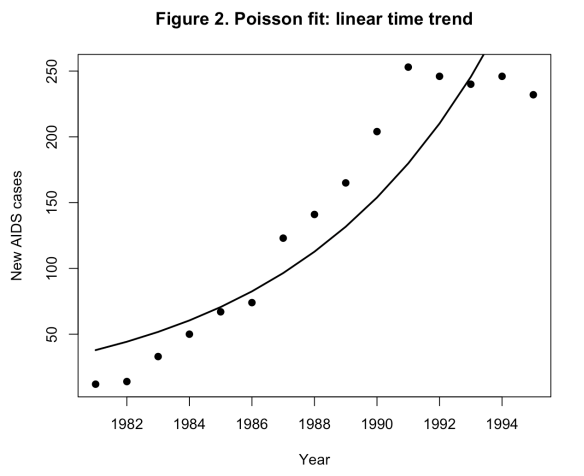
Figure 1 shows the number of new AIDS cases from 1981–1995. The number of cases increases rapidly during the 1980s, peaks in the early 1990s, and then begins to level off and slightly decline. This pattern suggests a nonlinear relationship between year and the number of new cases.



(b) Simple Poisson Model

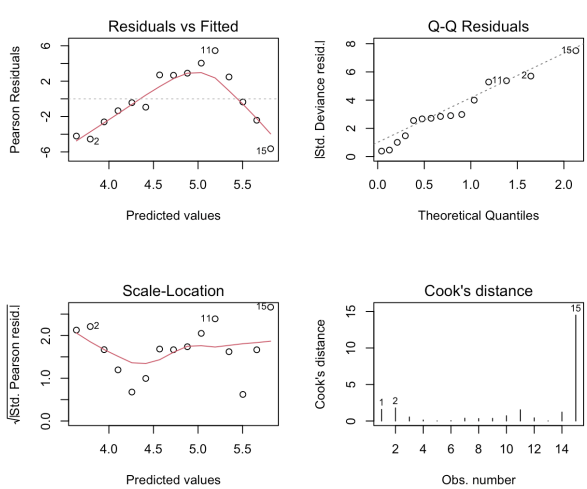
A Poisson regression model with $\log(\mu) = \beta 0 + \beta 1$ year was fit.

The fitted curve increases exponentially over time. However, the observed data plateau and decline after 1991, while the model continues increasing. This makes the simple log-linear model not adequately capture the curvature in the data.



(c) Residual Diagnostics

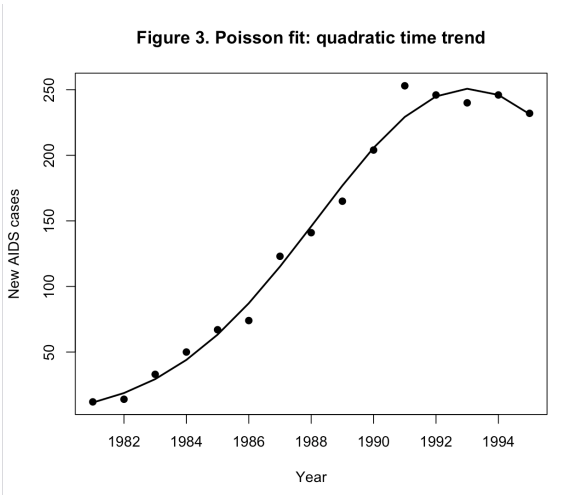
Residual plots reveal non-random patterns; this indicates model misspecification. The linear time trend fails to capture nonlinear behavior in the data. This suggests adding a nonlinear term, such as year^2.



(d) Quadratic Poisson Model

A quadratic term was added: $\log(\mu) = \beta_0 + \beta_1 \text{ year} + \beta_2 \text{ year}^2$

The new fitted curve now captures both the rapid increase and the leveling off. The quadratic model provides a substantially improved fit.



(e) Likelihood Ratio Test

Test statistic = **164.09**

Degrees of freedom = 1

p-value < **2.2 × 10^-16**

Since the p-value is extremely small, we reject the null hypothesis that the quadratic term is unnecessary. The quadratic Poisson model significantly improves model fit.

```
> anova(fit1, fit2, test="Chisq")
Analysis of Deviance Table

Model 1: cases ~ year
Model 2: cases ~ year + I(year^2)
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        13    173.335
2        12      9.245  1   164.09 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Question 2 – Cereal Shelf Placement (Multinomial Logistic Regression)

(a) Data Preparation

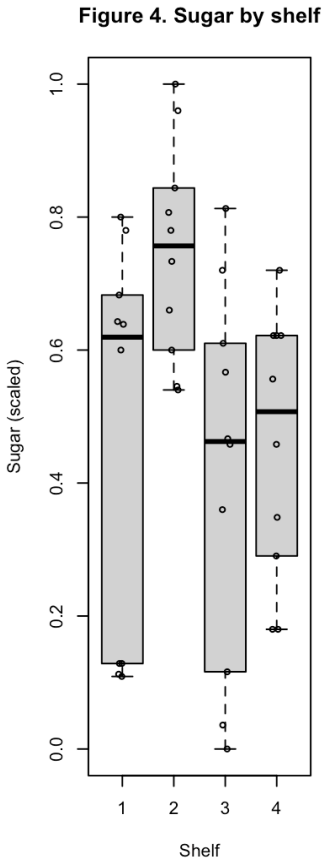
Sugar, fat, and sodium were standardized per gram and rescaled to the interval [0,1]. Shelf was treated as a categorical response variable.

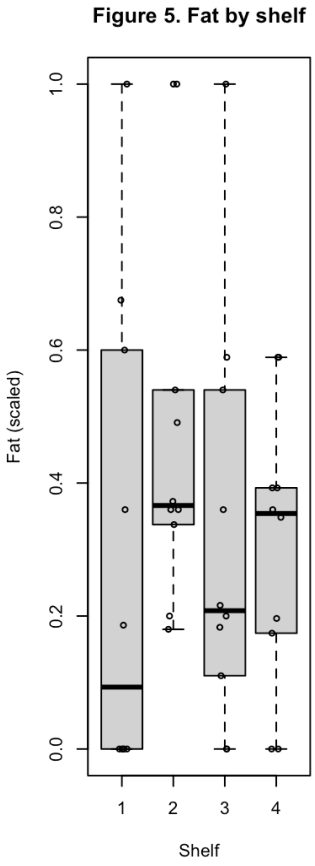
(b) Visual Analysis

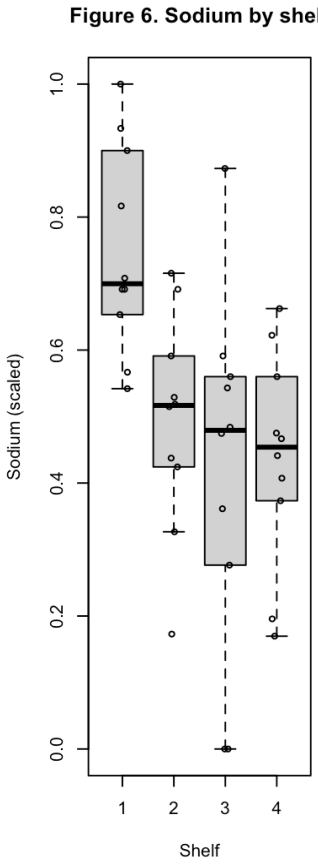
The boxplots reveal a noticeable difference in nutritional content across shelves.

Shelf 2 appears to contain cereals with higher sugar levels. Shelves 3 and 4 tend to have lower sugar values. The sodium variable also varies across shelves.

These visual differences suggest shelf placement is associated with nutritional content.







(c) Ordinality of Shelf

Although shelves 1–4 have a natural vertical order, the placement does not consistently increase or decrease as the predictor values change. Because the relationship between nutritional content and shelf does not increase or decrease uniformly, we should model Shelf as a nominal categorical variable using multinomial logistic regression.

(d) Most Important Predictor (AIC Comparison)

AIC values:

Full model = **91.19028**

No sugar = **107.95511**

No fat = **90.47384**

No sodium = **111.81001**

Removing sodium increases AIC the most. Therefore, sodium is the most important predictor, followed by sugar. Fat contributes the least to model performance.

```
> AIC(fit_full, fit_no_sugar, fit_no_fat, fit_no_sodium)
              df       AIC
fit_full      12  91.19028
fit_no_sugar   9 107.95511
fit_no_fat     9  90.47384
fit_no_sodium  9 111.81001
```

(e) Interpretation of Sugar Coefficients

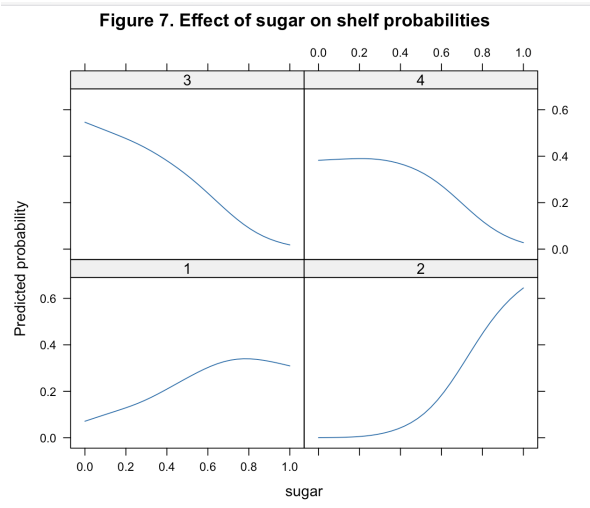
Relative to Shelf 1:

Shelf 2 sugar coefficient = **2.69**

Shelf 3 sugar coefficient = **−12.22**

Shelf 4 sugar coefficient = **−11.39**

Higher sugar increases the odds of placement on Shelf 2 and decreases the odds of placement on Shelves 3 and 4.



(f) Apple Jacks Prediction

Predicted probabilities:

Shelf 1 = **0.053**

Shelf 2 = **0.472**

Shelf 3 = **0.200**

Shelf 4 = **0.274**

Apple Jacks is most likely to be placed on Shelf 2.

(g) Best Shelf for Targeting Children

Shelf 2 appears most beneficial for targeting children. High-sugar cereals are strongly associated with Shelf 2, and children's cereals typically have higher sugar content.

Question 3 – Challenger O-Ring Failures (Logistic Regression)

(a) Independence Assumption

Logistic regression assumes independence among O-rings; but, O-rings within the same launch experience identical conditions, this may violate their independence. This could underestimate variability and lead to a too confident inference.

(b) Logistic Model (Temperature + Pressure)

Temperature coefficient = **−0.0983** (p = 0.0285)

Pressure coefficient not significant (p = 0.2691)

Lower temperatures increase the probability of O-ring failure.

```
> summary(fit_full)

Call:
glm(formula = cbind(O.ring, Number - O.ring) ~ Temp + Pressure,
    family = binomial, data = chall)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.520195   3.486784   0.723   0.4698
Temp        -0.098297   0.044890  -2.190   0.0285 *
Pressure     0.008484   0.007677   1.105   0.2691
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 24.230  on 22  degrees of freedom
Residual deviance: 16.546  on 20  degrees of freedom
AIC: 36.106

Number of Fisher Scoring iterations: 5
```

(c) Likelihood Ratio Test for Pressure

Deviance difference = **1.5407**

p-value = **0.2145**

At α = 0.1, we fail to reject the null hypothesis. Pressure does not significantly improve the model and may be removed.

```
> anova(fit_temp, fit_full, test="Chisq")
Analysis of Deviance Table

Model 1: cbind(O.ring, Number - O.ring) ~ Temp
Model 2: cbind(O.ring, Number - O.ring) ~ Temp + Pressure
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        21     18.086
2        20     16.546  1   1.5407   0.2145
>
```

(d) Model Simplification

The pressure variable was removed to simplify the model because it was not statistically significant. Statistical evidence does not support retaining Pressure, even though we could see omitted variable bias (OBV).

(e) Probability of Failure at 31°F

Predicted probability = **0.818**

There is a very high probability of O-ring failure at extremely low temperatures.

```
> predict(fit_temp, newdata=data.frame(Temp=31), type="response")
        1
0.8177744
```

(f) Probit Comparison

Logistic probability = **0.818**

Probit probability = **0.696**

Both models indicate high failure risk at low temperature, leading to the same conclusion.

```
> fit_temp_probit <- glm(cbind(O.ring, Number - O.ring) ~ Temp,
+                        data=chall, family=binomial(link="probit"))
>
> c(
+   logit  = predict(fit_temp, newdata=data.frame(Temp=31), type="response"),
+   probit = predict(fit_temp_probit, newdata=data.frame(Temp=31), type="response")
+ )
  logit.1  probit.1
0.8177744 0.6963991
```

Question 4 – Bikeshare Count Model (Negative Binomial Regression)

(a) Model Choice

Dispersion statistic from Poisson model = **25.60**

Because this value is greater than 1, there is a strong overdispersion.

AIC comparison:

Poisson AIC = **281,158.78**

Negative Binomial AIC = **89,491.41**

The Negative Binomial model provides a much better fit and is the appropriate model.

```
> disp
[1] 25.59892
```

```
> AIC(fit_pois, fit_nb)
         df       AIC
fit_pois 40 281158.78
fit_nb   41  89491.41
```

(b) Interpretation of Effects

Temperature IRR = **2.59**

A one-unit increase in temperature multiplies expected rentals by approximately 2.59.

Working day IRR = **0.842**

The rentals are about 16% lower on working days compared to non-working days.

Also, poor weather significantly reduces rentals.

Season (month) and hour of day have large and highly significant effects.

```
> summary(fit_nb)

Call:
glm.nb(formula = bikers ~ workingday + temp + weathersit + mnth +
    hr, data = bike, control = glm.control(maxit = 200), init.theta = 3.759869595,
    link = log)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                2.71388    0.03788  71.652  < 2e-16 ***
workingday1               -0.17255    0.01252 -13.786  < 2e-16 ***
temp                       0.95009    0.07215  13.169  < 2e-16 ***
weathersitcloudy/misty    -0.06433    0.01378  -4.669 3.03e-06 ***
weathersitlight rain/snow -0.57664    0.02122 -27.170  < 2e-16 ***
weathersitheavy rain/snow -0.82594    0.54310  -1.521    0.128
mnthFeb                    0.21184    0.03088   6.860 6.87e-12 ***
mnthMarch                  0.31791    0.03094  10.275  < 2e-16 ***
mnthApril                  0.61516    0.03551  17.324  < 2e-16 ***
mnthMay                    0.88679    0.04009  22.122  < 2e-16 ***
mnthJune                   0.84981    0.04636  18.331  < 2e-16 ***
mnthJuly                   0.76053    0.05011  15.176  < 2e-16 ***
mnthAug                    0.77729    0.04702  16.532  < 2e-16 ***
mnthSept                   0.87561    0.04200  20.849  < 2e-16 ***
mnthOct                    0.88435    0.03521  25.117  < 2e-16 ***
mnthNov                    0.78653    0.03285  23.945  < 2e-16 ***
mnthDec                    0.68371    0.03051  22.411  < 2e-16 ***
hr1                       -0.47519    0.04109 -11.566  < 2e-16 ***
hr2                       -0.80552    0.04194 -19.205  < 2e-16 ***
hr3                       -1.43093    0.04402 -32.503  < 2e-16 ***
hr4                       -2.02405    0.04705 -43.016  < 2e-16 ***
hr5                       -0.98889    0.04241 -23.319  < 2e-16 ***
hr6                        0.40625    0.04021  10.104  < 2e-16 ***
hr7                        1.41091    0.03969  35.547  < 2e-16 ***
hr8                        1.97342    0.03955  49.892  < 2e-16 ***
hr9                        1.47648    0.03966  37.225  < 2e-16 ***
hr10                       1.13254    0.03984  28.425  < 2e-16 ***
hr11                       1.27330    0.03992  31.900  < 2e-16 ***
hr12                       1.46818    0.04000  36.708  < 2e-16 ***
hr13                       1.46416    0.04014  36.479  < 2e-16 ***
hr14                       1.41613    0.04026  35.173  < 2e-16 ***
hr15                       1.44339    0.04028  35.833  < 2e-16 ***
hr16                       1.65911    0.04018  41.294  < 2e-16 ***
hr17                       2.11468    0.03995  52.936  < 2e-16 ***
hr18                       2.02939    0.03986  50.912  < 2e-16 ***
hr19                       1.71780    0.03976  43.208  < 2e-16 ***
hr20                       1.40905    0.03975  35.445  < 2e-16 ***
hr21                       1.14804    0.03975  28.881  < 2e-16 ***
hr22                       0.90393    0.03982  22.701  < 2e-16 ***
hr23                       0.51340    0.04001  12.832  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
(Dispersion parameter for Negative Binomial(3.7599) family taken to be 1)

    Null deviance: 40158.3  on 8644  degrees of freedom
Residual deviance:  9225.3  on 8605  degrees of freedom
AIC: 89491

Number of Fisher Scoring iterations: 1


              Theta:  3.7599
          Std. Err.:  0.0616

 2 x log-likelihood:  -89409.4070
```

(c) Most Important Predictors (ΔAIC)

Removing hr increases AIC by **11,497**, making it the most important predictor.

Removing mnth increases AIC by **1,100**, making it the second most important predictor.

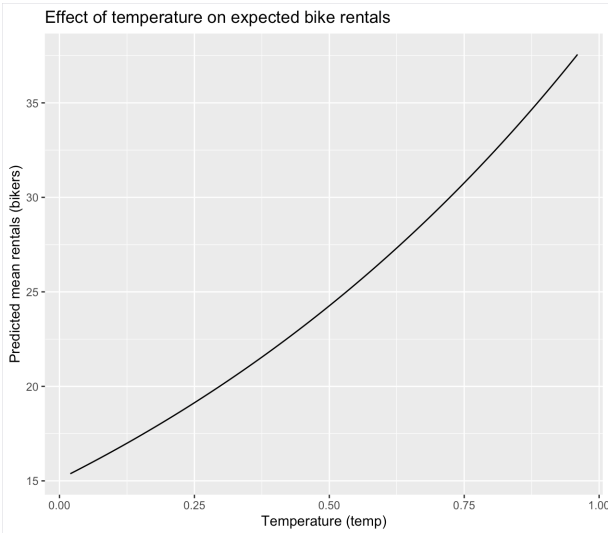
The hour of day drives the largest variation in bike rentals.

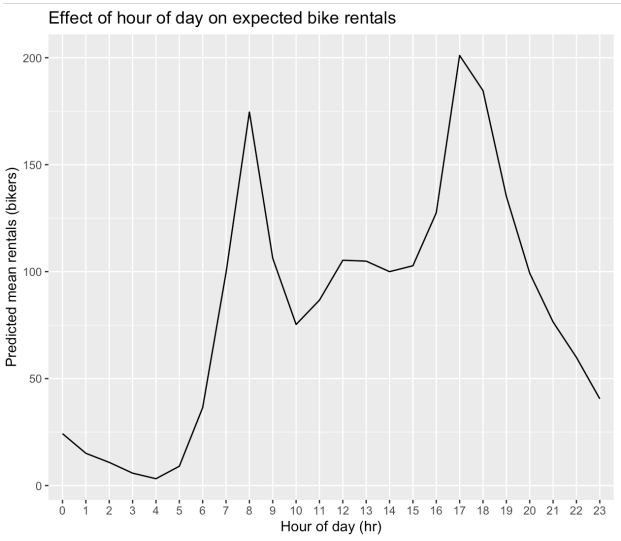
```
> imp_aic
          Model       AIC   DeltaAIC
1          Full  89491.41     0.0000
2 No workingday  89658.63   167.2190
3       No temp  89657.35   165.9395
4 No weathersit  90119.19   627.7777
5       No mnth  90591.64  1100.2278
6         No hr 100988.52 11497.1157
```

(d) Effect Plots

The temperature shows a steady increase in rentals as temperature rises.

The effect of the hour plot shows peaks around 8 AM and 5–6 PM, reflecting commuting behavior.





(e) Highest Rental Conditions

Highest rentals occur during:

- Warm temperatures
- Clear weather
- Summer months
- Late afternoon (5–6 PM)
- Non-working days

(f) Zero Inflation Check

Observed zero proportion = **0**

Model-implied zero probability = **0.0036**

Zero-inflated modeling is unnecessary.